Lab 4 — Working with Raster Data

Raster data, symbology, clipping and batch processing · QGIS 3.44 LTR

This lab follows the official **QGIS Training Manual, Module 7 — Rasters** (lessons 7.1 and 7.2) and two pages of the **QGIS User Manual**: **24.2.3.2 Clip raster by mask layer** and **23.6 The batch processing interface**. All screenshots are taken from QGIS with the lab data. They were made with QGIS 3.34; the steps and text follow the 3.44 documentation. Your dialogs may show a few extra options: leave them at their defaults. Steps marked **(lab addition)** are not in the manual; **(lab check)** and **(tested in this lab)** mark checks we added after testing the lab data.

Goals of this lab

- Understand what raster data is and how it differs from vector data.
- Load rasters, combine them into one virtual raster, reproject and merge them.
- Change the symbology and transparency of a raster.
- Clip a raster with a polygon (vector) layer.
- Run the same tool on many rasters at once (batch processing).

The data

- Unzip `Lab4_Data.zip`. It contains the folder `exercise_data` from the QGIS Training Manual data (`github.com/qgis/QGIS-Training-Data`).
- Put `exercise_data` in a path with English letters only and no spaces, e.g. `C:\GIS\exercise_data`.
- Save every result in the folder `exercise_data/output`.

From Lab 3 to Lab 4

	Vector (Lab 3)	Raster (this lab)
Made of	Features: points, lines, polygons	Pixels (a grid of cells)
Each item holds	Attributes in the attribute table	A value: a colour, an elevation, a temperature…
Clip	Vector clipped by vector	**Raster clipped by vector** (Part 3)

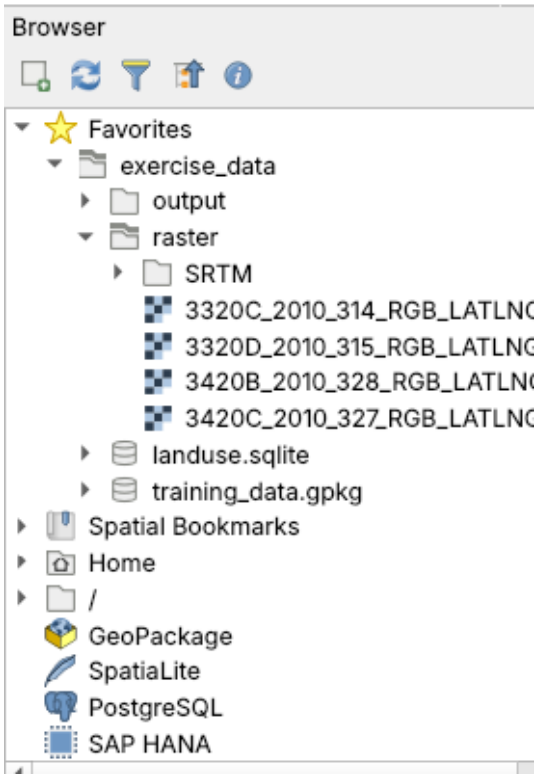
Training Manual 7.1: "Vector data has discrete features … Raster data, however, is like any image … objects … are represented using pixels with different values."

Part 1 · Working with Raster Data (Training Manual 7.1)

1. Loading Raster Data (7.1.1)

Raster data can be loaded with the same methods we used for vector data. The manual suggests the **Browser** panel.

1. **(lab addition, as in Lab 3)** In the **Browser** panel, right-click **Favorites → Add a Directory…** and choose your `exercise_data` folder.
2. Expand `exercise_data/raster`.
3. Load the four images (double-click or drag each one into the map): `3320C_2010_314_RGB_LATLNG.tif`, `3320D_2010_315_RGB_LATLNG.tif`, `3420B_2010_328_RGB_LATLNG.tif`, `3420C_2010_327_RGB_LATLNG.tif`.



The raster folder in the Browser panel



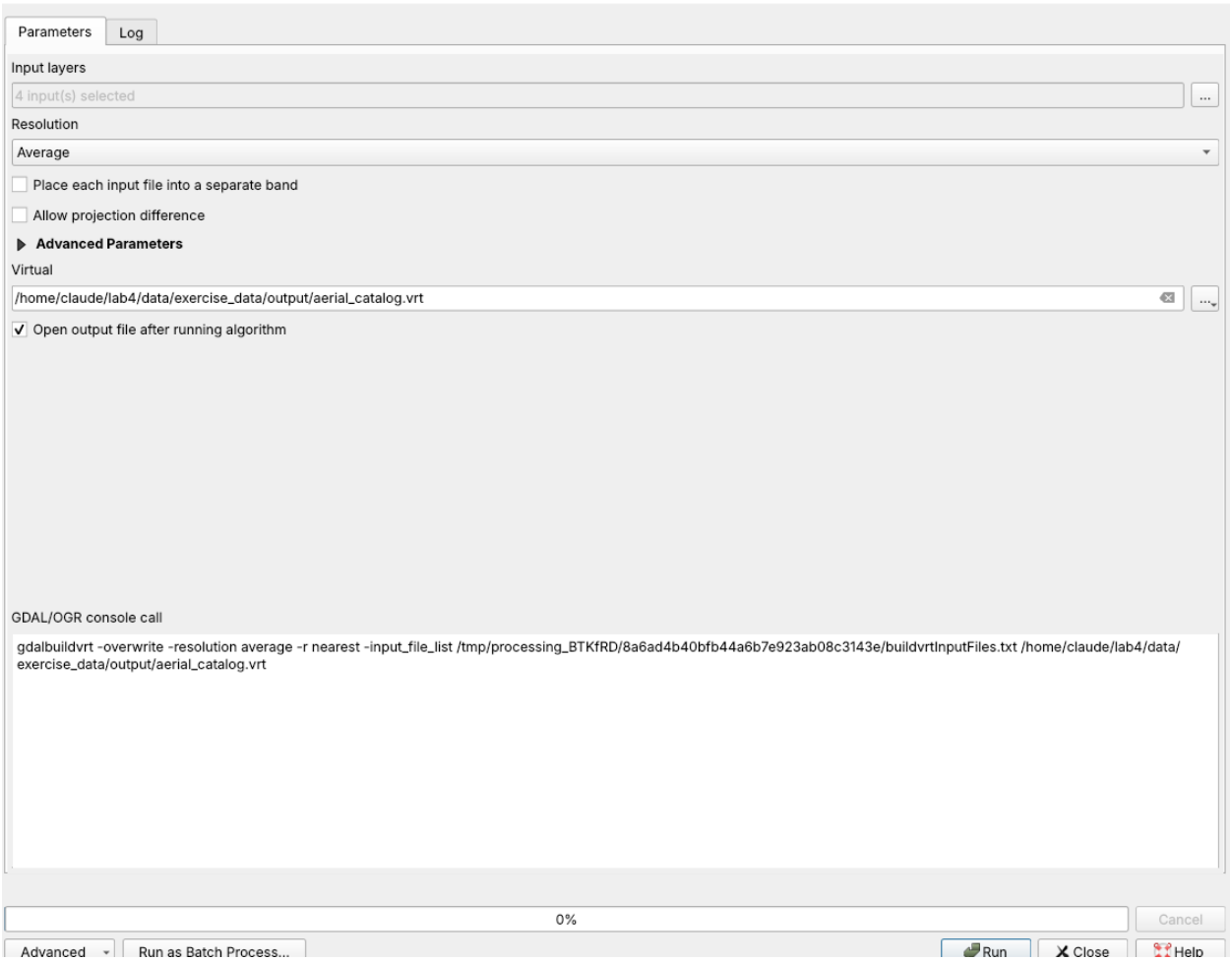
Four aerial images covering the study area

> ✔ **Check** 4 layers in the Layers panel. **(lab check)** Expand a layer in the Layers panel: each image has 3 bands, **Band 1 (Red)**, **Band 2 (Green)**, **Band 3 (Blue)**.

2. Create a Virtual Raster (7.1.2)

Working with four rasters all the time is not ideal. A **Virtual Raster** (also called a **Catalog**, i.e. a list of files) organises existing rasters into one file. It is **not** a new raster.

1. Open **Processing** → **Toolbox** and find **GDAL** → **Raster miscellaneous** → **Build virtual raster**.
2. Click **…** next to **Input layers** and check all four layers (or **Select All**).
3. **Uncheck Place each input file into a separate band**.
4. Look at the box **GDAL/OGR console call** at the bottom: this is the command that runs when you click **Run**. It changes when you change the options.
5. Processing creates temporary layers by default. To save the file, click **…** under **Virtual** → **Save to File…** → go to `exercise_data/output` → name it `aerial_catalog.vrt` **(lab addition)**.
6. Click **Run**.



Build virtual raster: 4 inputs, separate band unchecked, saved to output

You can now remove the four original rasters from the **Layers** panel and keep only the catalog.



One virtual raster (aerial_catalog) instead of four layers

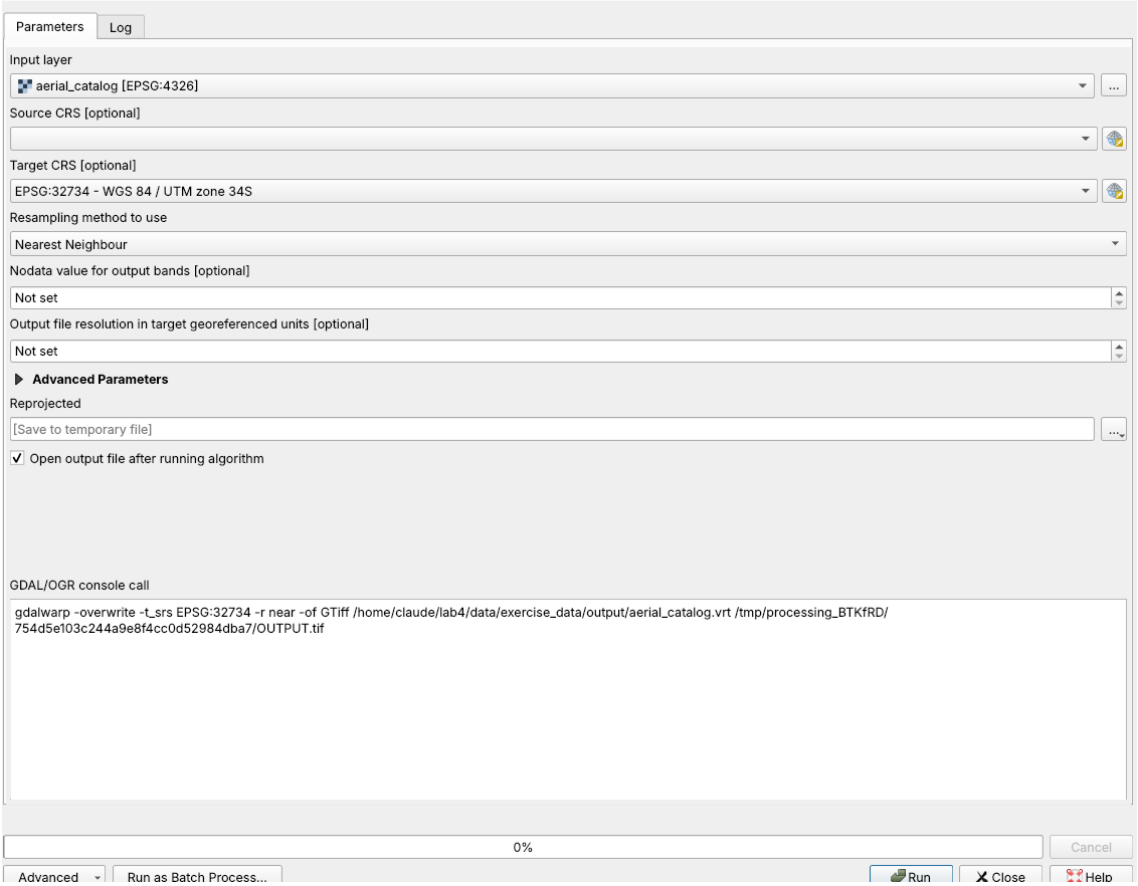
> ✓ **Check** One layer `aerial_catalog` shows all four images.

3. Transforming Raster Data (7.1.3)

Look at the two dialogs in this section only; you do not need to click Run. A catalog merges and reprojects “on the fly” (while drawing, without making a new file). If you will use the data for a long time, it can be more efficient to create new rasters that are **already merged and reprojected**. This improves performance, but takes time to set up.

Reprojecting rasters

Open **GDAL → Raster projections → Warp (reproject)**. You can also reproject a virtual raster. Example **(lab addition)**: **Input layer** `aerial_catalog`, **Target CRS** `EPSG:32734 – WGS 84 / UTM zone 34S` (the CRS we used in Lab 3). Note: the DEM in Part 2 uses a different zone, `EPSG:32733` (UTM 33S).

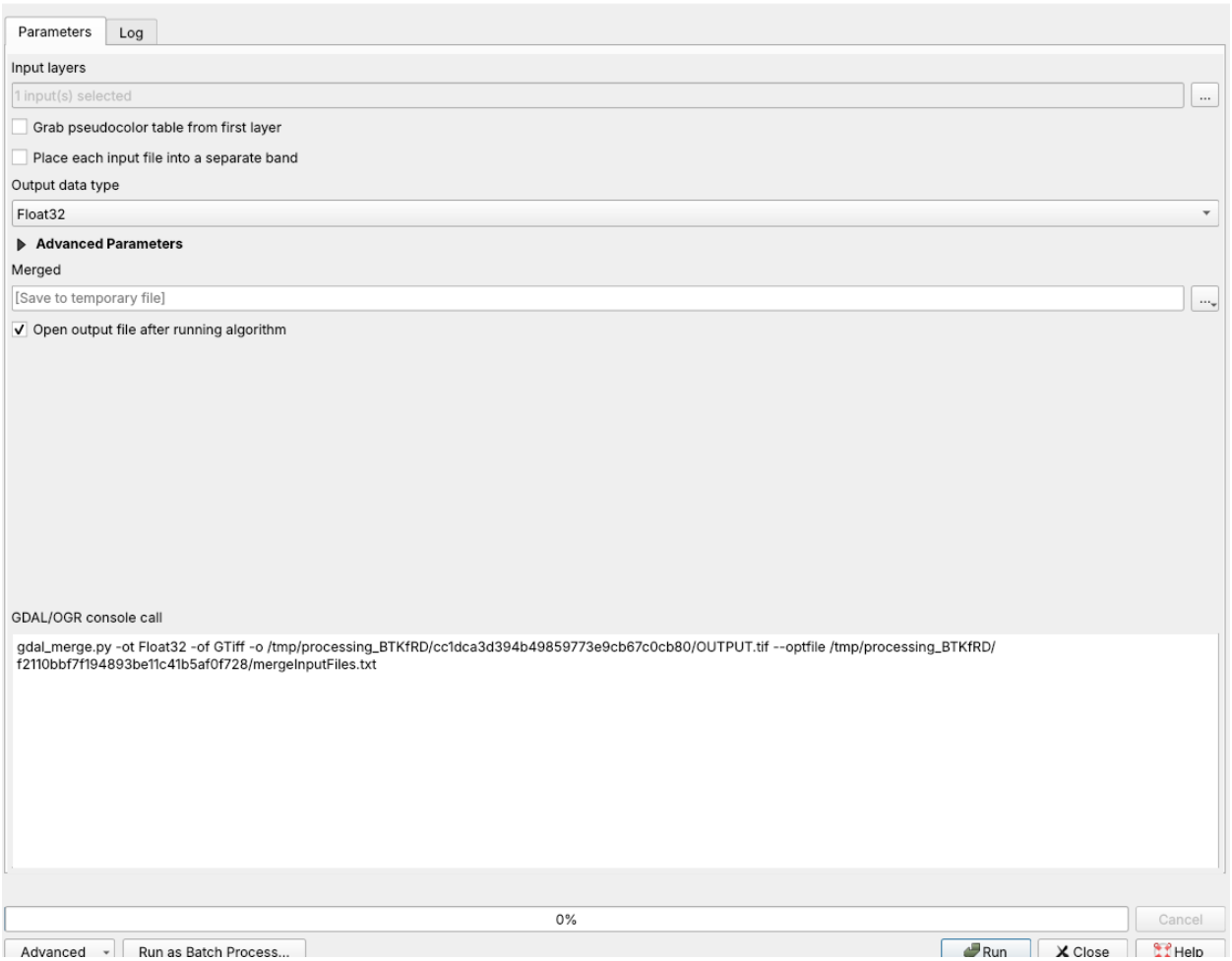


Warp (reproject) with Target CRS EPSG:32734

Merging rasters

To create a new raster and save it to disk, use **GDAL → Raster miscellaneous → Merge**. Choose the layers with the **…** button. A virtual raster can also be the input: then all rasters inside it are processed.

> **Note (Training Manual)**
>
> - Depending on how many rasters you merge and their resolution, the new file can be **really big**. Consider a virtual raster (section 2) instead.



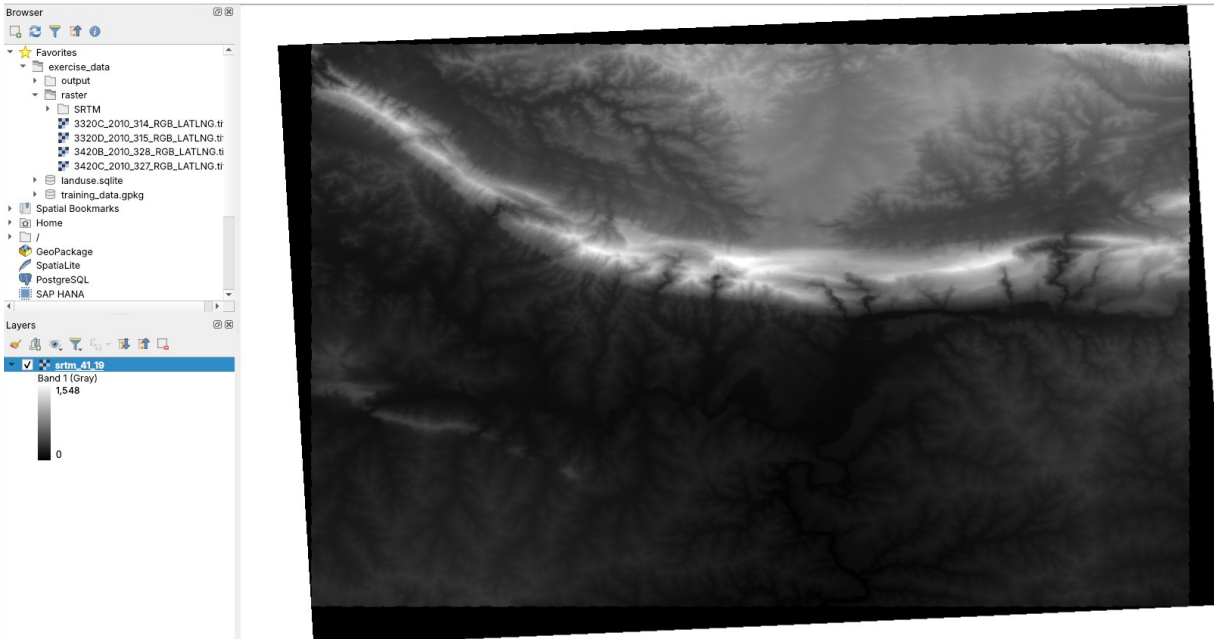
Merge with the virtual raster as input

Part 2 · Changing Raster Symbology (Training Manual 7.2)

4. Load a DEM

1. Remove `aerial_catalog` from the Layers panel (optional).
2. From the Browser, load `exercise_data/raster/SRTM/srtm_41_19.tif`.
3. Right-click it in the Layers panel → **Zoom to Layer(s)** (the manual calls it **Zoom to Layer**).

This dataset is a **Digital Elevation Model (DEM)**: a map of the elevation of the terrain. In the aerial images each pixel held a **colour**; in a DEM each pixel holds an **elevation value**. QGIS shows it in grey and stretches the values automatically (it spreads them between black and white).



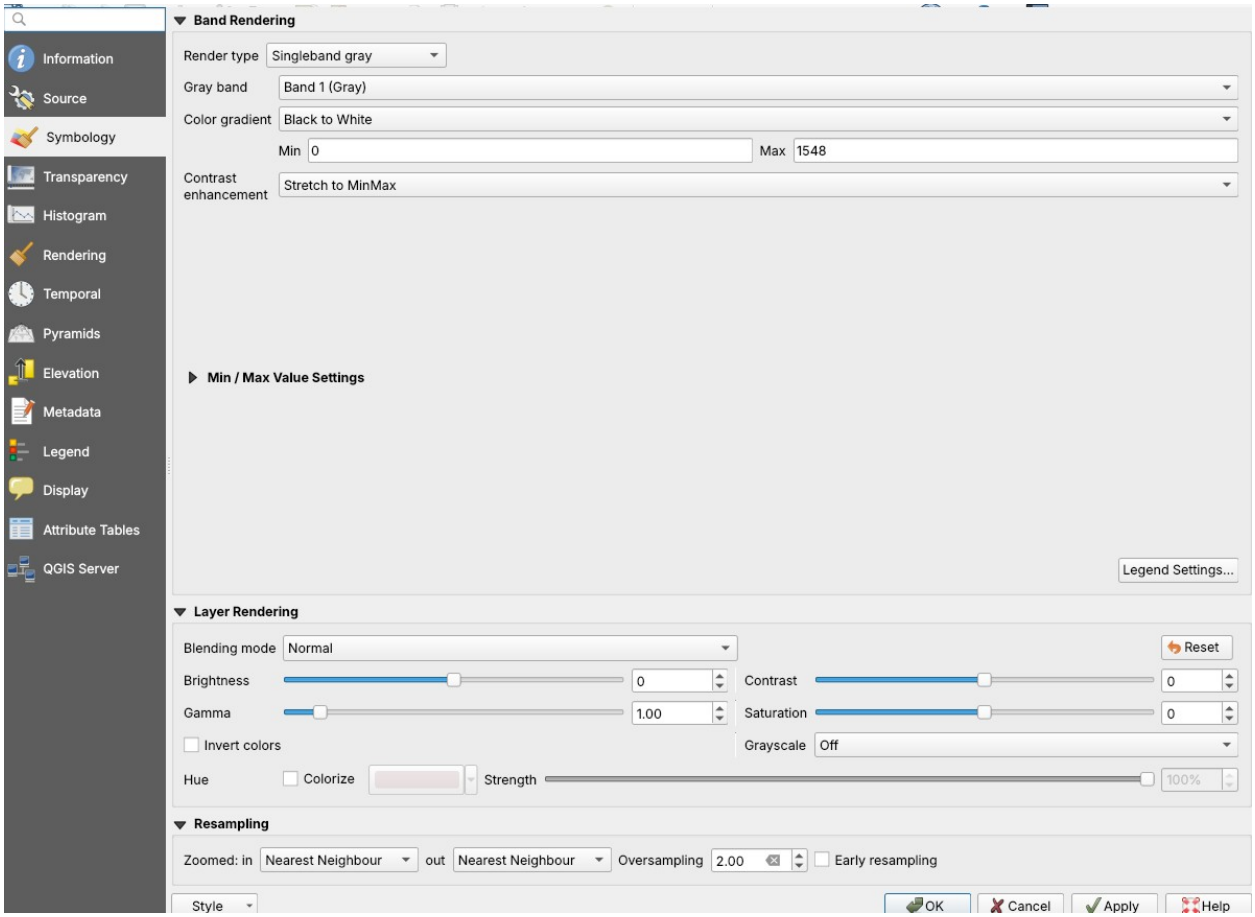
srtm_41_19 shown in grey (default)

5. Two ways to change the symbology

- Right-click the layer → **Properties** → **Symbology** tab, or
- Click **Open the Layer Styling panel** above the Layers panel (shortcut **F7**) → **Symbology**.

6. Singleband gray

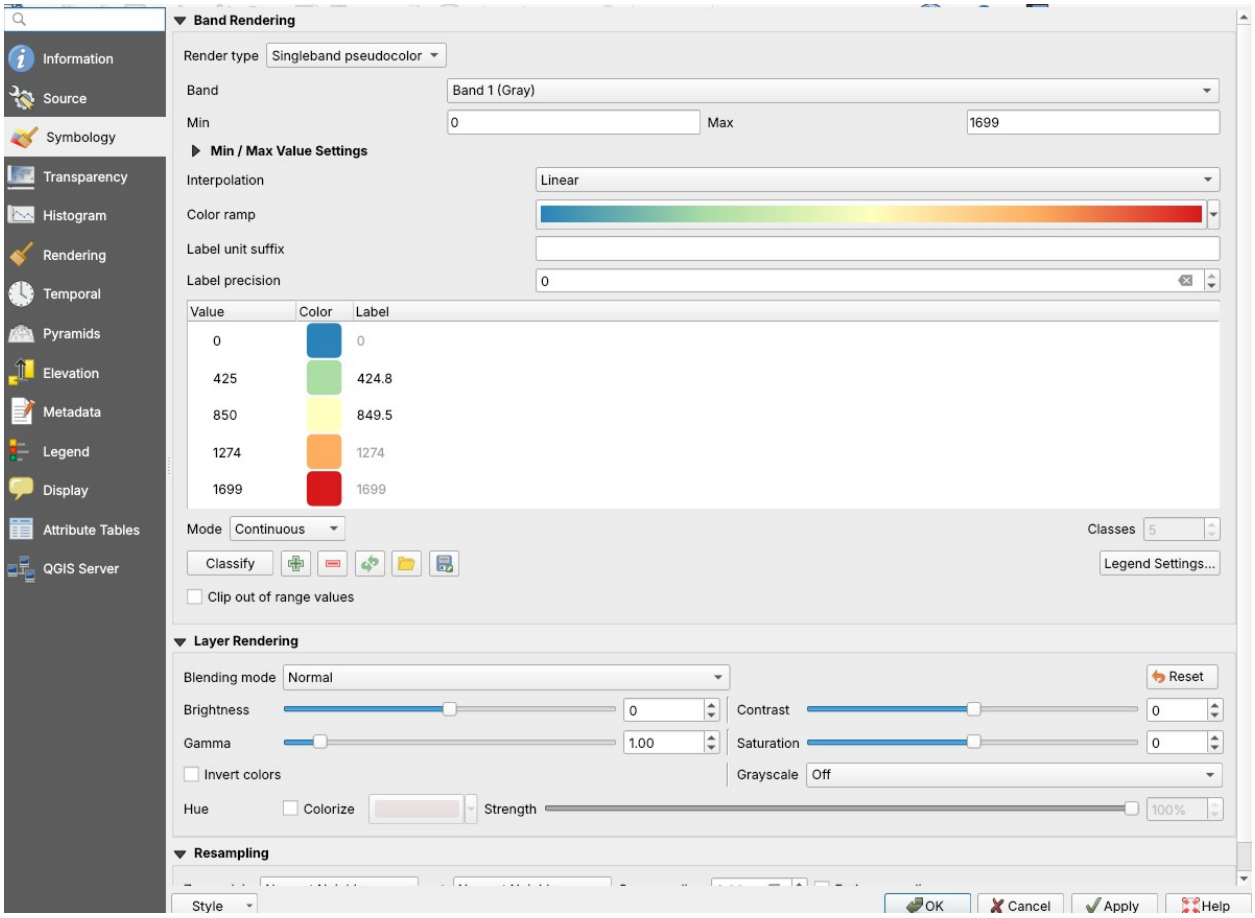
Setting	Meaning
Color gradient	`Black to white`: low values black, high values white. Try `White to black`.
Contrast enhancement	`Stretch to MinMax`: pixel values are stretched between the minimum and the maximum.
Min / Max Value Settings	How min and max are found: **User Defined** (you type them) · **Cumulative count cut** (cuts e.g. 2% extreme values) · **Min / max** (Accuracy: Actual or Estimate) · **Mean +/- standard deviation**.



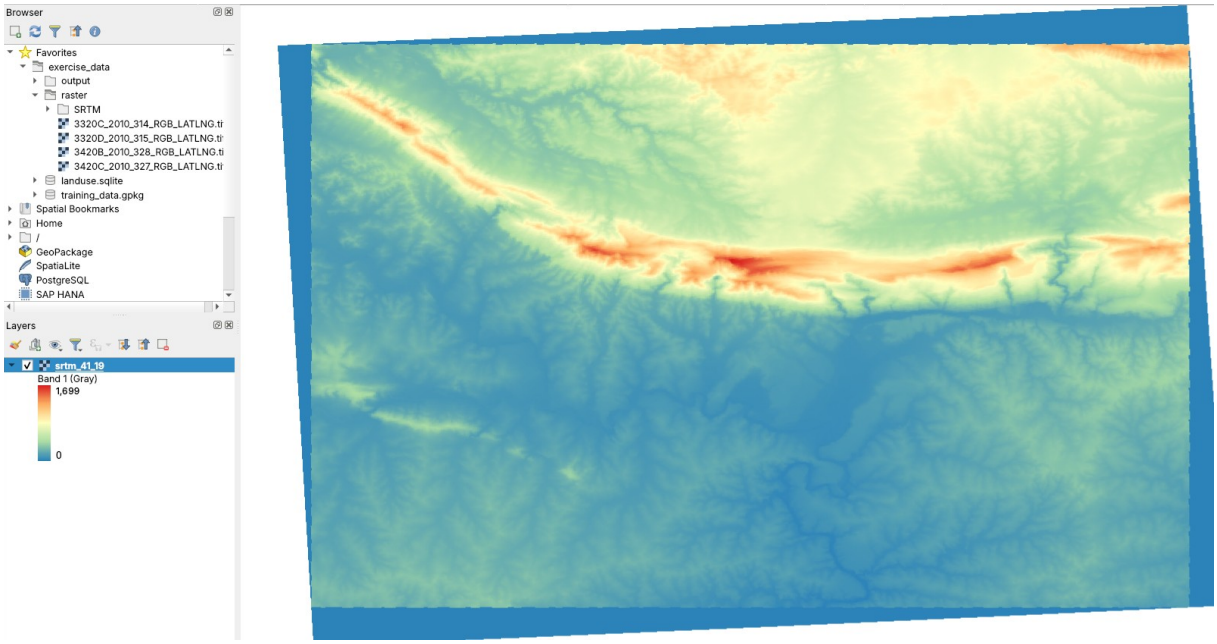
Symbology tab: Singleband gray

7. Singleband pseudocolor

1. Change **Render type** to **Singleband pseudocolor**.
2. If you don't like the default colours, choose another **Color ramp**. **(lab addition)** In the screenshots: `Spectral`, then **Invert Color Ramp** from the ramp's drop-down, so low = blue and high = red.
3. **(lab addition)** Open **Min / Max Value Settings** → choose **Min / max**, Accuracy **Actual**, so the colours use the full range 0 – 1,699 (the default cuts the extreme 2 %).
4. Click **Classify**, then **OK**.



Singleband pseudocolor after Classify



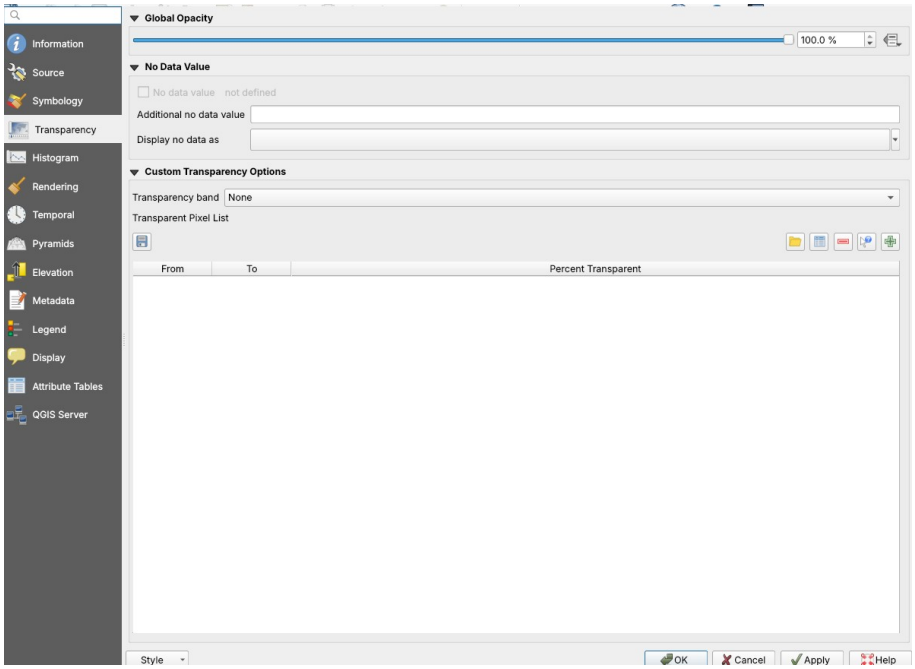
The DEM from blue (low) to red (high)

✓ Check Low areas (valleys) are blue, high areas (mountains) are red.

8. Changing the transparency

Whole layer

Properties → **Transparency** tab → move the **Global Opacity** slider down. This helps to see layers under the raster.

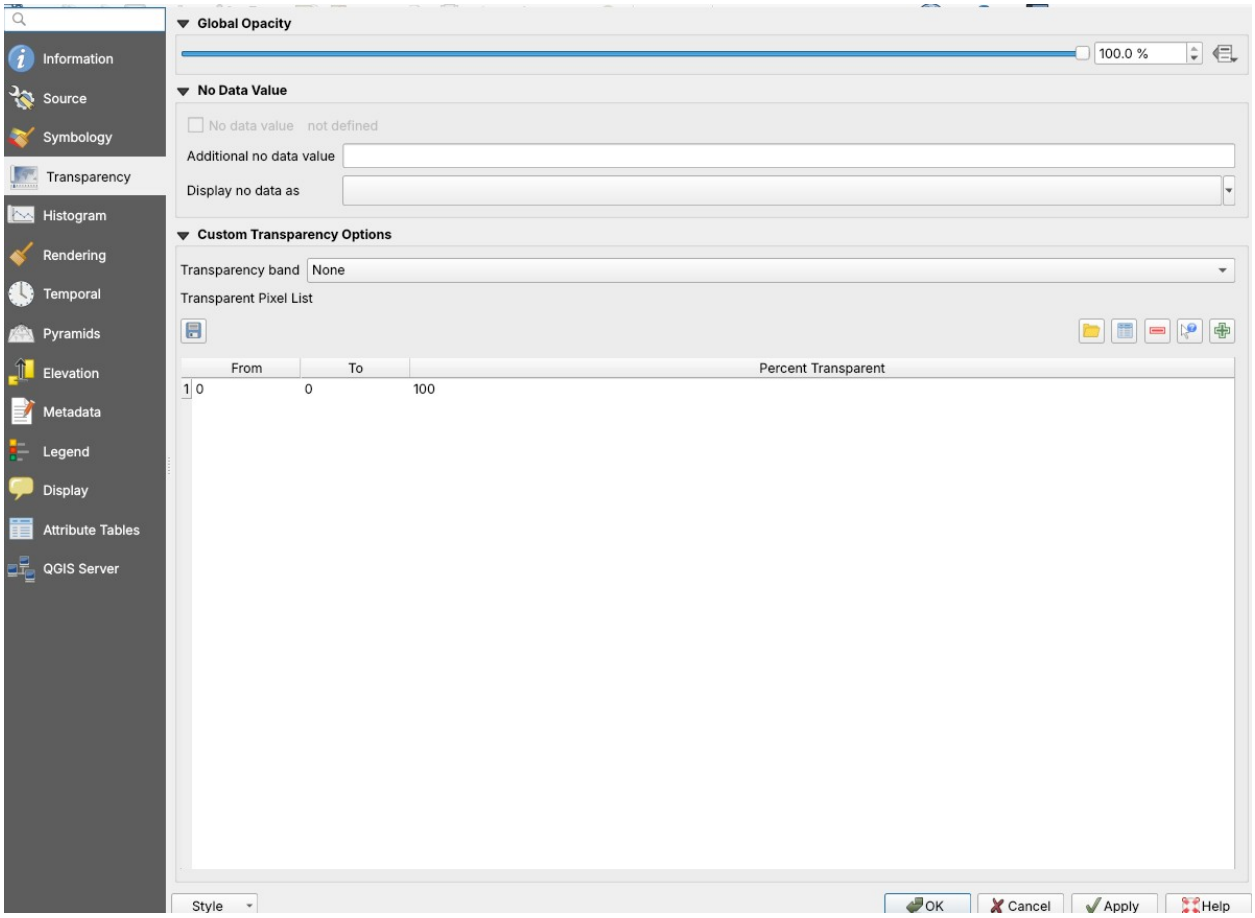


Transparency tab: Global Opacity (before changing, 100 %)

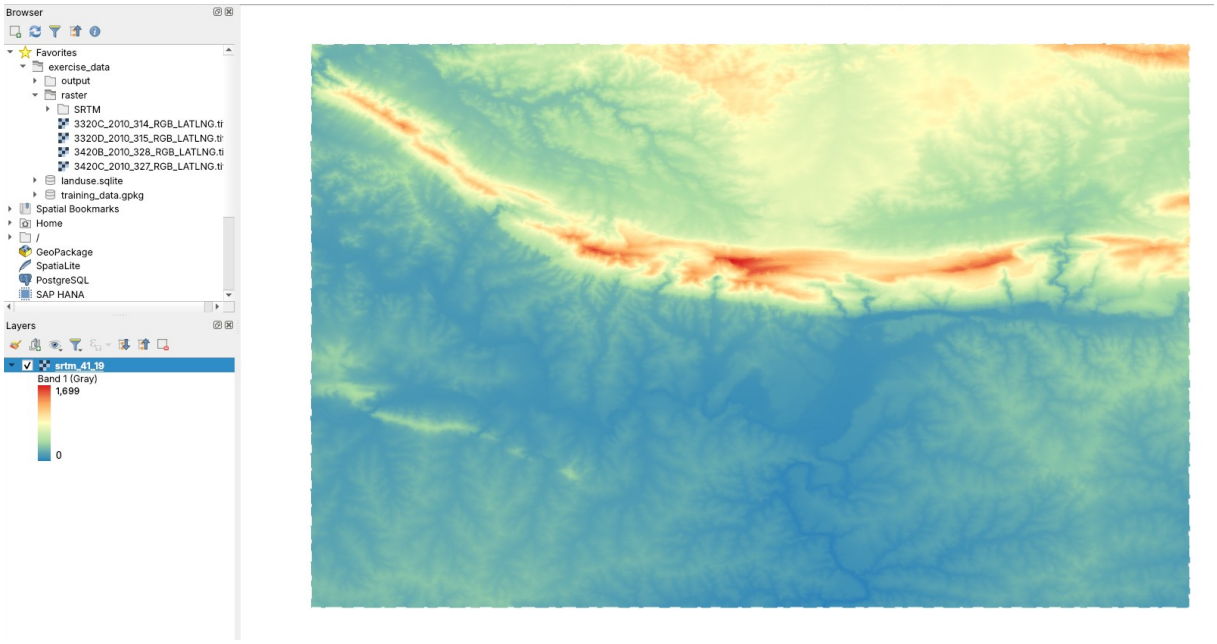
Some pixel values

Set **Global Opacity** back to 100 % first, so you can see the effect clearly. The DEM has a homogeneous colour (one single colour) in the corners. To make these pixels transparent:

1. In the **Transparency** tab, under **Custom Transparency Options**, click the **Add values from display** icon (arrow with a question mark). The dialog disappears.
2. Click on the homogeneous colour in a corner of the DEM. The table is filled with the clicked value (here **0**, 100 % transparent).
3. Click **OK**.



Transparent Pixel List: value 0 → 100 % transparent



The corners are now transparent

(**Add values manually** adds a range of values instead.)

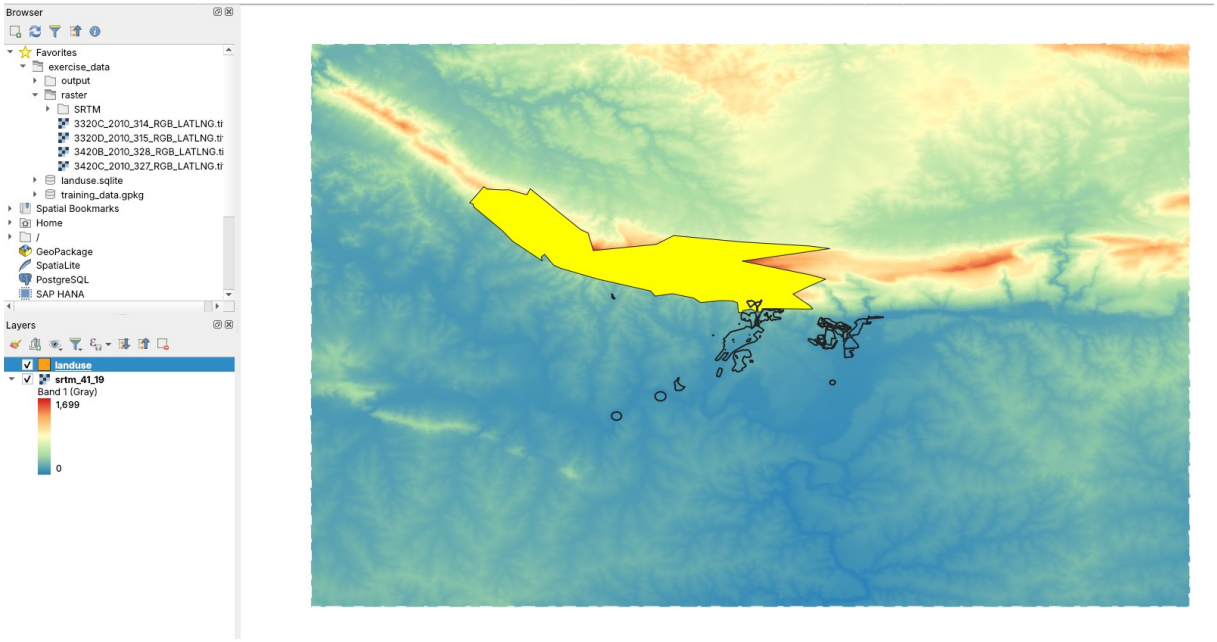
Part 3 · Clip raster by mask layer (User Manual 24.2.3.2)

This tool “clips any GDAL-supported raster by a vector mask layer”. The mask must be a **polygon** layer. Default menu (User Manual): **Raster → Extraction**; the menu item is **Clip Raster by Mask Layer…**.

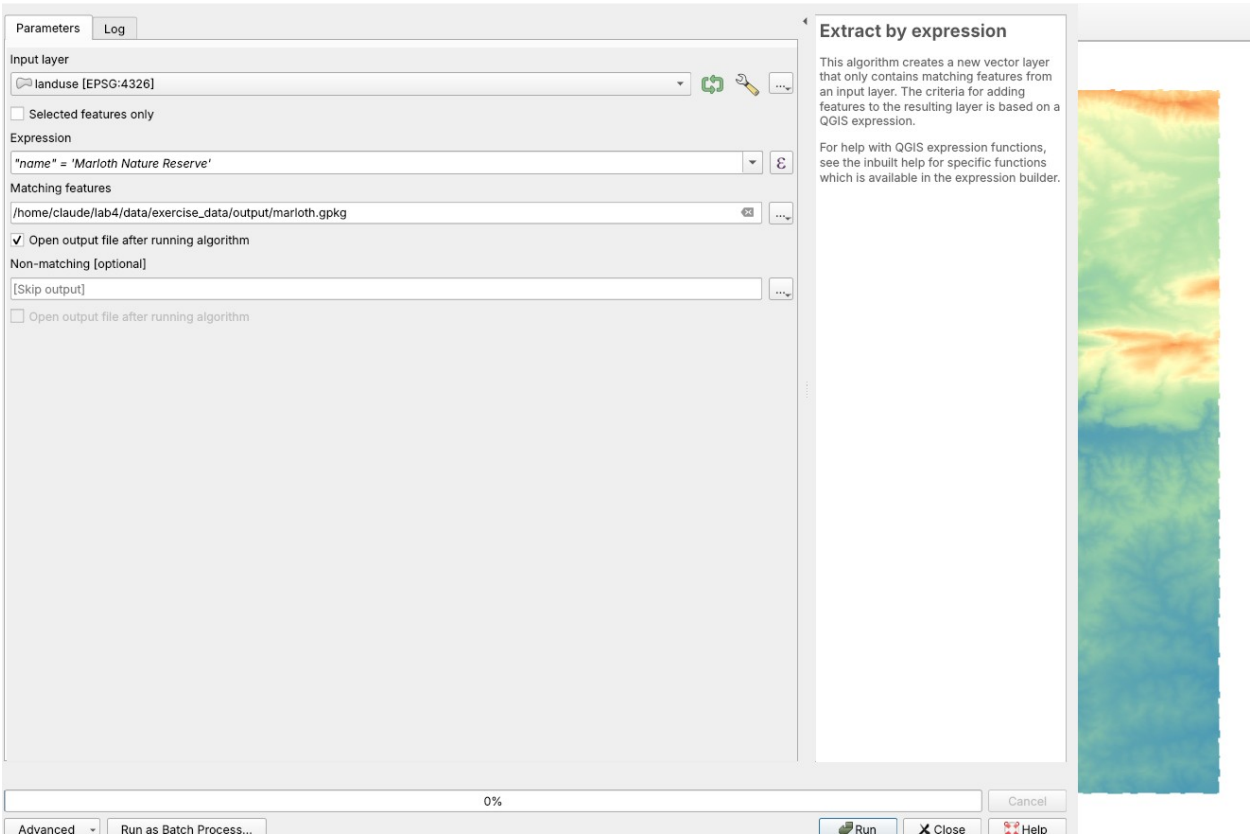
9. Prepare the mask (a boundary polygon)

We clip the DEM to the boundary of the **Marloth Nature Reserve (lab addition)** (a polygon in `landuse.sqlite`). First we save that polygon as its own layer, with **Extract by expression** from Lab 3.

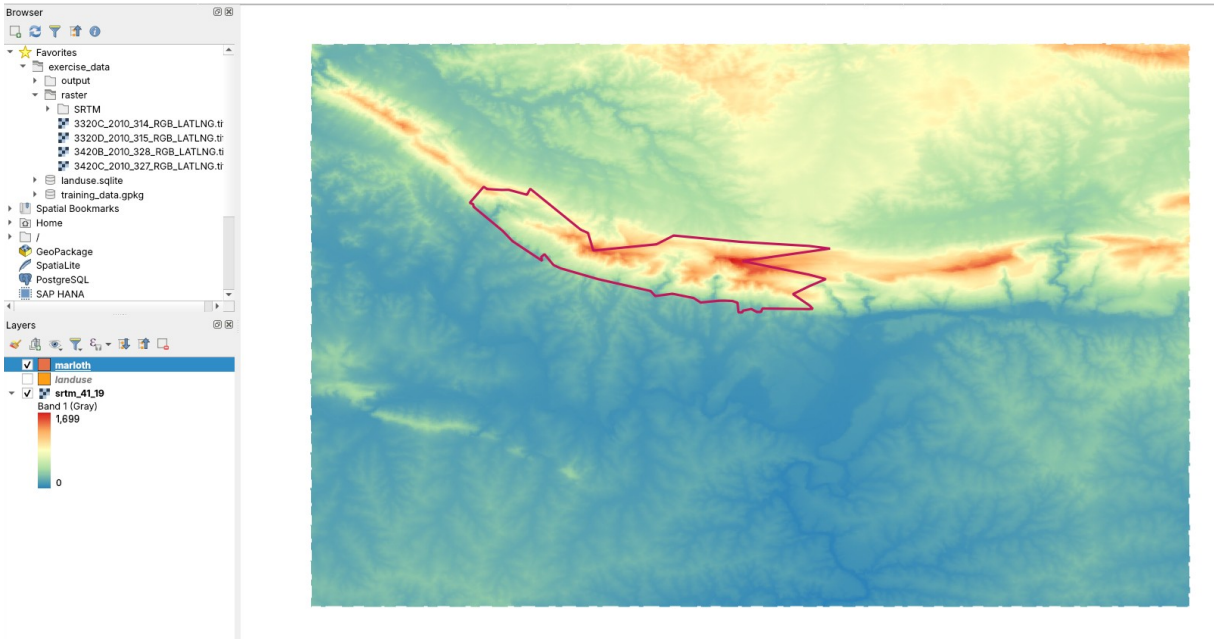
1. Load `exercise_data/landuse.sqlite` (layer `landuse`).
2. (Optional, before extracting: **Select Features by Expression** on `landuse` with the expression below — the polygon turns yellow.)
3. **Processing Toolbox → Vector selection → Extract by expression**. **Input layer**: `landuse`. **Expression**: `"name" = 'Marloth Nature Reserve'`.
4. **Matching features**: click **…** → **Save to File…** → `exercise_data/output/marloth.gpkg` → **Run**.



Optional preview: the Marloth polygon selected (yellow) over the DEM



Extract by expression: name = 'Marloth Nature Reserve'



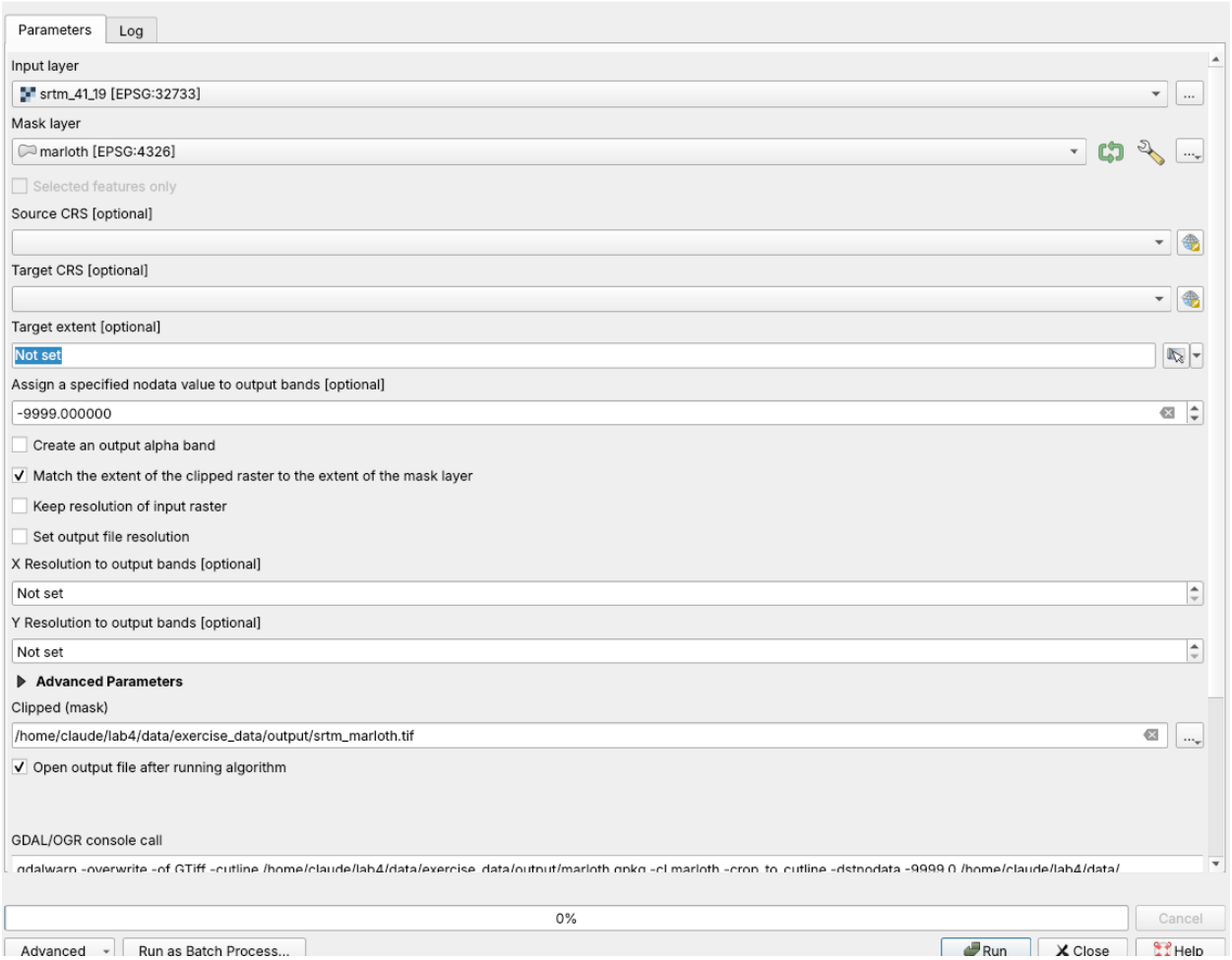
The new layer marloth (1 polygon)

✔ **Check** `marloth` has **1** feature.

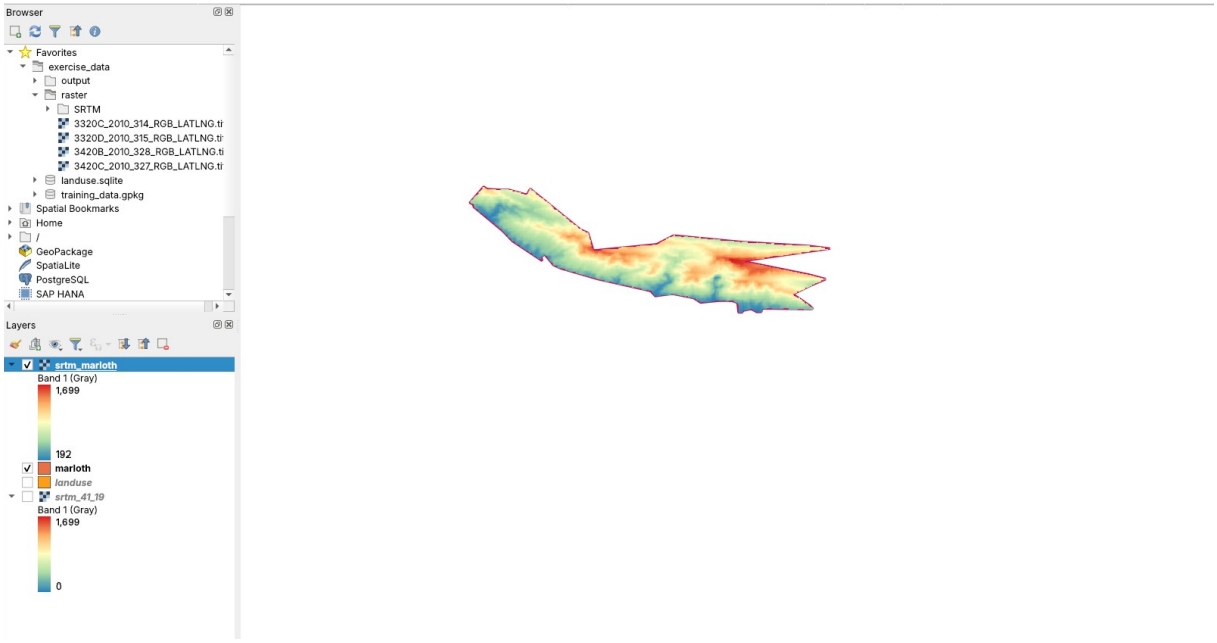
10. Clip the DEM

1. **Raster → Extraction → Clip Raster by Mask Layer**.
2. Fill in the parameters as in the table, then **Run**.

Parameter	Value	Why
Input layer	`srtm_41_19`	The raster to clip
Mask layer	`marloth`	The polygon used to clip
Assign a specified nodata value to output bands (optional)	`-9999`	Pixels outside the polygon get this value, so QGIS does not draw them **(lab addition)**
Match the extent of the clipped raster to the extent of the mask layer	checked (default)	The output covers only the extent (bounding box) of the polygon
Clipped (mask)	**…** → **Save to File…** → `output/srtm_marloth.tif`	Save the result



Clip raster by mask layer: input srtm_41_19, mask marloth, NoData -9999



Result: the DEM only inside the reserve

✓ **Check** Only the DEM inside the reserve is drawn, with no frame around it.

Why NoData? (tested in this lab)

- Without a NoData value, the pixels outside the polygon get the value **0**, and they are drawn as a coloured frame around the result.

Part 4 · Batch processing (User Manual 23.6)

Every algorithm can run as a **batch process**: the same tool runs many times, once for each set of inputs. This is useful for large amounts of data. Here we clip **all** rasters in `raster/SRTM` with the same mask **(lab addition)**.

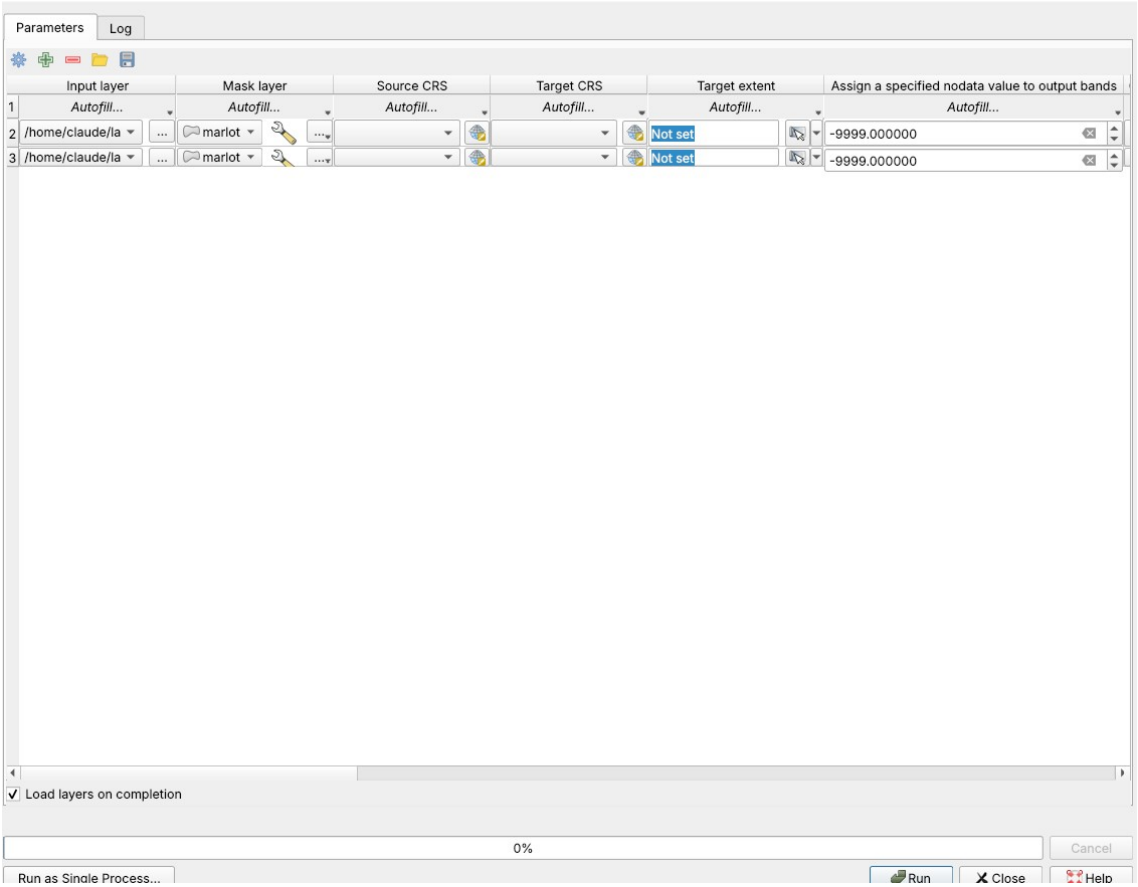
11. Open the batch interface

- In the Toolbox, right-click the algorithm → **Execute as Batch Process…**, or
- In the algorithm dialog, click **Run as Batch Process…**.

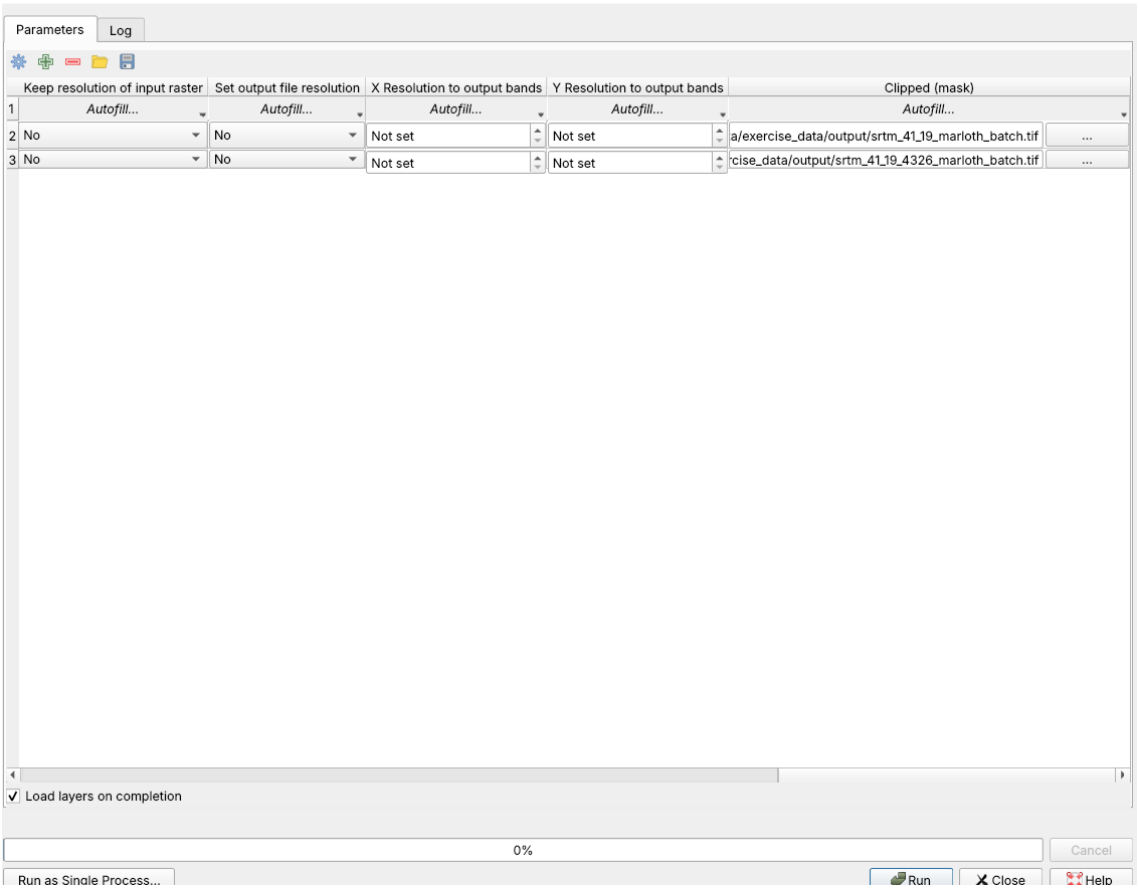
The dialog is a table: **each row is one run**, **each column is one parameter**. The first row has **Autofill…** menus to fill the cells below.

12. Fill the table

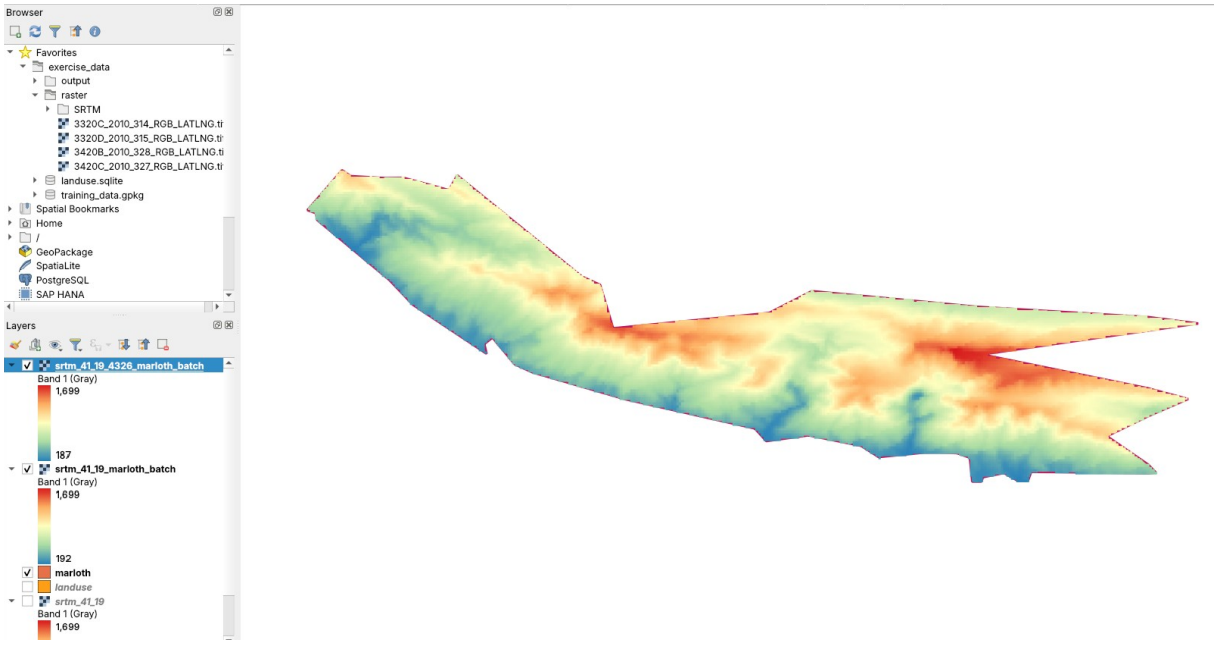
1. **Input layer** → **Autofill…** → **Add Files by Pattern…**. **Look in**: `exercise_data/raster/SRTM`. **File pattern**: `*.tif`. Rows are added for `srtm_41_19.tif` and `srtm_41_19_4326.tif`. If an empty row is left, select it (click its number) and click **Remove row(s)**.
2. **Mask layer**: choose `marloth` in the first data row (row 2; row 1 is the Autofill row) → open **Autofill…** at the top of the column → **Fill Down**.
3. **Assign a specified nodata value**: type `-9999` in row 2 → **Autofill…** of that column → **Fill Down**.
4. **Clipped (mask)**: type the output path in each row, e.g. `C:\GIS\exercise_data\output\srtm_41_19_marloth_batch.tif` and `C:\GIS\exercise_data\output\srtm_41_19_4326_marloth_batch.tif` (use your own folder) (or click **…** → **Select File/Folder…**). Tip: after choosing a file in the first row, **Fill with parameter values** → **Input layer** adds the input name to the end of the names in the rows below.
5. Tick **Load layers on completion** at the bottom → **Run**.



Batch table (left part): 2 inputs, mask marloth, NoData -9999



Batch table (right part): one output file per row; Load layers on completion ticked



Both rasters clipped in one run

✓ Check Two new layers, one for each input file, both cut to the reserve.

Save the batch

- The **Save** button in the batch toolbar saves the table as a `.json` file, and **Open** loads it again — useful to repeat the same batch later.

Summary

Task	Tool
Load a raster	Browser panel
Several rasters as one	GDAL → Build virtual raster
Change CRS / join rasters	GDAL → Warp (reproject) / Merge
Colour a raster	Symbology: Singleband gray / pseudocolor
Hide pixels	Transparency: Global Opacity / Custom Transparency
Cut a raster with a polygon	Raster → Extraction → Clip Raster by Mask Layer
Same tool on many files	Run as Batch Process…

References

- Training Manual 7.1 — Working with Raster Data — https://docs.qgis.org/3.44/en/docs/training_manual/rasters/data_manipulation.html
- Training Manual 7.2 — Changing Raster Symbology — https://docs.qgis.org/3.44/en/docs/training_manual/rasters/changing_symbology.html
- User Manual 24.2.3.2 — Clip raster by mask layer — https://docs.qgis.org/3.44/en/docs/user_manual/processing_algs/gdal/rasterextraction.html#clip-raster-by-mask-layer
- User Manual 23.6 — The batch processing interface — https://docs.qgis.org/3.44/en/docs/user_manual/processing/batch.html